

Recognizing and Managing Postpartum Psychosis: A Clinical Guide for Obstetric Providers.

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Postpartum psychosis is a psychiatric emergency that affects 1 to 2 per 1000 women. Key clinical features include mood fluctuation, abnormal thoughts or behaviors, and confusion. Women with a history of bipolar disorder are at heightened risk, as are first-time mothers; current research on the causes focuses on biological triggers, such as immune dysregulation. Women with postpartum psychosis require inpatient hospitalization and should be treated with lithium, antipsychotics, and benzodiazepines.

Genetic and Epigenetic Factors Associated with Postpartum Psychosis: A 5-Year Systematic Review.

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Purpose: Postpartum psychosis (PPP) is a serious mental health illness affecting women post-parturition. Around 1 in 1000 women are affected by postpartum psychosis, and the symptoms usually appear within 2 weeks after birth. Postpartum mental disorders are classified into 3 main categories starting from the least to most severe types, including baby blues, postpartum depression, and postpartum psychosis. **Materials and Methods:** In this systematic review, genetic and epigenetic factors associated with postpartum psychosis are discussed. A PRISMA flow diagram was followed, and the following databases were used as main sources: PubMed, ScienceDirect, and Scopus. Additional information was retrieved from external sources and organizations. The time period for the articles extracted was 5 years. **Results:** Initially, a total of 2379 articles were found. After the stated criteria were applied, 58 articles were identified along with 20 articles from additional sources, which were then narrowed down to a final total of 29 articles. **Conclusions:** It can be concluded that there is an association between PPP and genetic and epigenetic risk factors. However, based on the data retrieved and examined, the association was found to be greater for genetic factors. Additionally, the presence of bipolar disorder and disruption of the circadian cycle played a crucial role in the development of PPP.

India's Silent Struggle: A Scoping Review on Postpartum Depression in the Land of a Billion Mothers.

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Postpartum emotional issues, including postpartum blues, postpartum psychosis, and postpartum depression (PPD), affect many mothers worldwide. The prevalence of postpartum blues ranges from 300 to 750 per 1,000 mothers, while postpartum psychosis has a prevalence of 0.89-2.6 cases per 1,000 births. PPD affects 100-150 new mothers per 1,000 births, and its prevalence can go above 25% globally. Depression in mothers can have long-lasting impacts on children and the next generation, affecting their growth, development, and relationships. However, the condition of PPD is frequently misdiagnosed, and women who suffer from it do not receive treatment. There is a lack of solid systematic evidence that looks at the overall burden of PPD and the risk factors that are associated with it in India. The awareness of PPD is critical in managing the symptoms and seeking professional support, and studies are needed to evaluate and assess the situation in India. This study is a systematic literature review with a scoping

review approach aimed at providing existing evidence on the awareness and prevalence of PPD in India. The search strategy for the electronic bibliographic search included a list of related terms and factors of awareness and prevalence of PPD. The search encompassed six prominent databases. The assessment tools used in studies from 2012 to 2023 study the factors using the Postpartum Depression Literacy Scale (PoDLiS) and the Edinburgh Postpartum Depression Scale, which measure the awareness and prevalence of PPD. The study found a significantly low volume of studies conducted in India to determine the prevalence of PPD. The materials or studies accessible are hospital-based research with a minimal number of samples. The reported prevalence rate of postpartum depression (PPD) in India varies across different studies, ranging from 4% to 48.5%. Among mothers who gave birth either naturally or by cesarean section in a hospital setting, one study reported the lowest prevalence rate (4%). The prevalence of PPD among women in India is estimated to be 18% based on the average score across all research. A study that evaluated the moms in the community found that PPD was prevalent in new mothers at a rate of 21.5%. There are three studies, in particular, that focus on how aware people are of PPD. Two studies found that 54% of the family members of postpartum moms and women undergoing tertiary pediatric care are aware of PPD. Many studies point to the lack of social support as a significant risk factor for mental health complications after delivery. History of depression, anxiety, adverse life events, partner abuse, low economic status, complications at childbirth, and a traumatic childbirth experience are some of the associated factors identified in the studies. Studies have also shown that women who undergo a cesarean section have a higher risk of developing PPD than those with a vaginal delivery.

Porcine maternal infanticide as a model for puerperal psychosis.

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Childbirth is a period of substantial rapid biological and psychological change and a wide range of psychotic disorders can occur ranging from mild 'baby blues' to severe episodes of psychotic illnesses. Puerperal psychosis is the most extreme form of postnatal psychosis, occurring in 1 in 1,000 births. In this study, we have used the pig as an animal model for human postnatal psychiatric illness. Our aim was to identify quantitative trait loci (QTL) associated with maternal (infanticide) sow aggression. This is defined by sows attacking and killing their own newborn offspring, within 24 hr of birth. An affected sib pair whole genome linkage analysis was carried out with 80 microsatellite markers covering the 18 porcine autosomes and the X chromosome, with the aim of identifying chromosomal regions responsible for this abnormal behavior. Analysis was carried out using the non-parametric linkage test of Whittemore and Halpern, as implemented in the Merlin software. The results identified 4 QTL mapping on Sus scrofa chromosomes 2 (SSC2), 10 (SSC10), and X (SSCX). The peak regions of these QTL are syntenic to HSA 5q14.3-15, 1q32, Xpter-Xp2.1, and Xq2.4-Xqter, respectively. Several potential candidate genes lie in these regions in addition to relevant abnormal behavioral QTL, found in humans and rodents.

Puerperal psychosis mimicking eclampsia.

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Puerperal psychosis occurs after delivery in 1-2/1,000 births. It usually presents after delivery, however, it also may present in the antepartum period. We report the third case which presented prior to delivery without a preceding history of maternal puerperal psychosis and the first which presented with catatonia

and symptoms of eclampsia. Although uncommon, the first presentation of psychosis during pregnancy should be considered part of the differential diagnosis in pregnant patients presenting with altered mental status after organic causes are excluded.

Prenatal depression leading to postpartum psychosis.

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Postpartum psychosis is a mood disorder occurring up to 3 months after delivery. Incidence is one to two women every 1,000 live births. If not detected and appropriately treated in time, it may have detrimental effects on both the mother and her baby. We report a case of puerperal psychosis in a patient with a history of depression. We have also reviewed the relevant literature discussing prediction, management and differential diagnosis of postpartum psychosis. We emphasise the importance of early detection and provision of care to all women at risk of mental illness by multidisciplinary team, including GPs, obstetricians, midwives and perinatal mental health professionals.

Obstetric Characteristics and Management of Patients with Postpartum Psychosis in a Tertiary Hospital Setting.

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Background. Postpartum psychosis is the most severe and uncommon form of postnatal affective illness. It constitutes a medical emergency. Acute management emphasizes hospitalization to ensure safety, antipsychotic medication adherence, and treatment of the underlying disorder. **Objective.** The aim of the study was to determine the obstetric characteristics and management of patients with postpartum psychosis in a tertiary centre in North-Western Nigeria. **Methodology.** This was a 10-year retrospective study. Records of the patients diagnosed with postpartum psychosis from January 1st, 2002, to December 31st, 2011, were retrieved and relevant data extracted and analyzed using the SPSS for Windows version 16.0. **Results.** There were 29 cases of postpartum psychosis giving an incidence of 1.1 per 1000 deliveries. The mean age of the patients was 20.6 ± 4 years. Twelve (55%) were primiparae, 16 (72.7%) were unbooked, and 13 (59%) delivered at home. All had vaginal deliveries at term. There were 12 (52.2%) live births, and 11 (47.8%) perinatal deaths and the fetal sex ratio was equal. The most common presentation was talking irrationally. **Conclusion.** There is need for risk factor evaluation for puerperal psychosis during the antenatal period especially in primigravidae and more advocacies to encourage women to book for antenatal care in our environment.

Perturbation and resilience of the gut microbiome up to 3 months after β -lactams exposure in healthy volunteers suggest an important role of microbial β -lactamases.

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Antibiotics notoriously perturb the gut microbiota. We treated healthy volunteers either with cefotaxime or ceftriaxone for 3 days, and collected in each subject 12 faecal samples up to day 90. Using untargeted and targeted phenotypic and genotypic approaches, we studied the changes in the bacterial, phage and

fungal components of the microbiota as well as the metabolome and the β -lactamase activity of the stools. This allowed assessing their degrees of perturbation and resilience. While only two subjects had detectable concentrations of antibiotics in their faeces, suggesting important antibiotic degradation in the gut, the intravenous treatment perturbed very significantly the bacterial and phage microbiota, as well as the composition of the metabolome. In contrast, treatment impact was relatively low on the fungal microbiota. At the end of the surveillance period, we found evidence of resilience across the gut system since most components returned to a state like the initial one, even if the structure of the bacterial microbiota changed and the dynamics of the different components over time were rarely correlated. The observed richness of the antibiotic resistance genes repertoire was significantly reduced up to day 30, while a significant increase in the relative abundance of β -lactamase encoding genes was observed up to day 10, consistent with a concomitant increase in the β -lactamase activity of the microbiota. The level of β -lactamase activity at baseline was positively associated with the resilience of the metabolome content of the stools. In healthy adults, antibiotics perturb many components of the microbiota, which return close to the baseline state within 30 days. These data suggest an important role of endogenous β -lactamase-producing anaerobes in protecting the functions of the microbiota by de-activating the antibiotics reaching the colon. Video Abstract.

β -Lactam Biotransformations Activate Innate Immunity.

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Antibiotics are widely prescribed to treat bacterial infections, but many of these drugs also affect patient immune responses. While the molecular mechanisms regulating these diverse immunomodulatory interactions are largely unknown, recent studies support two primary models: (1) antibiotics can alter immune function by directly interacting with human targets; and/or (2) antibiotics can indirectly affect immune responses via alteration of the human microbiota composition. Here, we describe results that could support a third model in which a nonimmunostimulatory antibiotic can be biotransformed by human microbiota members into an immunostimulatory product that lacks antibacterial activity. Specifically, we identified, characterized, and semisynthesized new biotransformation products derived from the β -lactams amoxicillin and ampicillin, antibiotics regularly prescribed in the clinic. The drug metabolism products were identified in bacterial cultures harboring β -lactamase, a common resistance determinant. One of the amoxicillin biotransformation products activated innate immunity, as assessed by NF- κ B signaling in human leukemic monocytes, whereas amoxicillin itself exhibited no effect. Amoxicillin has previously been shown to have minimal long-term impact on human microbiota composition in clinical trial studies. Taken together, our results could support a broader immunomodulatory mechanism whereby antibiotics could indirectly regulate immune function in a stable, microbiome-dependent manner.

Low dose oral beta-lactamase protects the gut microbiome from oral beta-lactam-mediated damage in dogs.

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Antibiotics, while lifesaving, damage the gut microbiome and can precipitate proliferation of pathobionts. A strategy to preserve gut microbiome integrity is to eliminate biologically active antimicrobials excreted

into the gastrointestinal tract (GI) without negatively affecting antibiotic therapeutic efficacy. Clinical proof of concept was achieved with SYN-004 (ribaxamase), a beta-lactamase enzyme formulated for oral delivery with intravenous penicillins and cephalosporins. Ribaxamase inactivated intestinal ceftriaxone, protected the gut microbiome, and significantly reduced the incidence of *Clostridioides difficile* disease. For use with oral beta-lactam antibiotics, a delayed release formulation of ribaxamase, SYN-007, was engineered for dissolution in the lower small intestine distal to the site of oral antibiotic absorption. In dogs that received oral amoxicillin, SYN-007 reduced microbiome disruption without interfering with amoxicillin systemic absorption. Here, a study to determine the lowest effective dose of SYN-007 was performed. Dogs received amoxicillin (40 mg/kg, PO, TID) +/- SYN-007 (PO, TID) at three doses, 10 mg, 3 mg, or 1 mg for five days. Serum amoxicillin levels, measured after the first and last antibiotic doses, were not significantly different +/-SYN-007 at all dose levels indicating that SYN-007 did not interfere with amoxicillin systemic absorption. Microbiome analyses demonstrated that amoxicillin significantly reduced bacteria richness and microbiome diversity resulting in altered microbiome composition. However, with all doses of SYN-007, microbiome richness and diversity were not significantly different from pretreatment and changes in microbiome composition were attenuated. These data demonstrate that effective SYN-007 doses can be reduced at least 10-fold while maintaining gut microbiome preservation. The potential to employ low SYN-007 doses to protect the gut microbiota has important implications for enhancing therapeutic outcomes for patients receiving oral beta-lactam antibiotics while simultaneously reducing cost per dose and ultimately, healthcare expenses.